# Lecture Notes:
# Sophisticated Hand Ciphers 2
# Double Transposition

Math F113X

# Double Transposition Cipher

- Use two keywords (often of different lengths).
- Do a transposition cipher with the first keyword to produce a first ciphertext.
- If there are spaces, keep track of the columns they're in. (Don't add junk letters!)
- Then encrypt that ciphertext (as though it were plaintext) using the second keyword.

# Encryption

1. Write the plaintext in rows.
2. Permute the columns using the keyword.
3. Read off the **columns** to form the new “plaintext”
4. Write it in rows in the second grid.
5. Permute the columns using the second keyword. Keep the spaces where they go.
6. Read off the **columns** to form the final ciphertext.

# Encryption

**Example:** Encrypt the word TRANSPOSITION using the first key `KEY` and the second key `SHORT`.

| K | E | Y |
|---|---|---|
| T | R | A |
| N | S | P |
| O | S | I |
| T | I | O |
| N | | |

reorder ⟼ Columns

| E | K | Y |
|---|---|---|
| R | T | A |
| S | N | P |
| S | O | I |
| I | T | O |
| N | | |

read out ⟼ Columns

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| S | H | O | R | T |
|---|---|---|---|---|
| R | S | S | I | T |
| N | O | T | N | A |
| P | I | O | | |

→

| H | O | R | S | T |
|---|---|---|---|---|
| S | S | I | R | T |
| O | T | N | N | A |
| I | O | | P | |

⟼ SOIST OINRN PTA

# Decryption

1. Count the number of characters. Determine how many “extra” characters there will be for the second keyword.
2. Write your second grid with the **sorted** keyword.
3. Fill in the columns of the grid, making sure to put the extra characters in the appropriate columns.
4. (Un)sort the columns
5. Read off the **rows**. ...

# Decryption, cont.

6. Using the number of characters, determine the number of “extra” characters for the first keyword.
7. Fill in the columns of the grid, filling in the extra long columns using the **sorted** keyword.
8. Unsort the columns.
9. Read off the **rows**

# Example: Decryption

Undo our encryption of the word TRANSPOSITION using the first kev KEY and the second key SHORT.

SOIST OINRN PTA ← ciphertext

13 characters

$\frac{13}{5} = 2 \text{ R } 3$

3 extra chara.

| * | * | | * | |
|---|---|---|---|---|
| H | O | R | S | T |
| S | S | I | R | T |
| O | T | N | N | A |
| I | O | | P | |

⟼

| * | * | * | | |
|---|---|---|---|---|
| S | H | O | R | T |
| R | S | S | I | T |
| N | O | T | N | A |
| P | I | O | | |

SHORT ↦ HORST

KEY ↦ EKY

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$\frac{13}{3} = 4 \text{ R } 1$

| | * | |
|---|---|---|
| E | K | Y |
| R | T | A |
| S | N | P |
| S | O | I |
| I | T | O |
| | N | |

⟼

| * | | |
|---|---|---|
| K | E | Y |
| T | R | A |
| N | S | P |
| O | S | I |
| T | I | O |
| N | | |

TRANSPOSITION

# Example: Decryption

Decrypt the ciphertext ETETS PIHXC R if it was encrypted using sing the first key `KEY` and the second key `SHORT`.

